Longer by how much?

The note inside. · Curiosity day

Name _____ Date _____



A comparison model, not two measured piano strings. Compare length while imagining the material, thickness, and tension stay the same.

PRETEND DRAWING LENGTHS · CENTIMETERS

Line	A	B
Length (cm)	24	9

These are invented drawing lengths. Do not measure the painting or real piano strings.

1. Line A is 24 cm long. Line B is 9 cm long. How much longer is A?

24 − 9 = _____

Show your thinking.

2. Imagine drawing A and B end to end. What is their total length?

24 + 9 = _____

Show your thinking.

3. Little erases 6 cm from line A. How long is that line now?

24 − 6 = _____

Show your thinking.

Sketch two bars that start at the same place. Show which part is the extra 15 cm. A sketch need not be actual size.

WONDER DAILY

The note inside.

The note inside



One leaf. Six steps. A story that keeps going.

GRADE 2

One key sinks beneath Little's hand. A note rings through the museum. "Was it hiding in there?" Little asks. Big listens beside the bench. The key hardly moved. Something inside the piano must have done much more.

An enormous model gives them a peek. A key tips; connected parts lift a felt-covered hammer. Tap! The hammer strikes a string and comes away. "A tiny hammer?" Little whispers. This team of moving parts is called the action — the key's small press becomes a tap.

Make a little book.

1. Print one-sided: US Letter, portrait, actual size / 100%. No fit. Headers/footers off.
2. Fold long edge to long edge; reopen. Fold short edge to short edge, then in half again. Open: eight boxes.
3. Cut only the solid center line across the middle two boxes. Keep the outer boxes joined.
4. Refold long edge to long edge, print outside. Push ends together to open a diamond, then a cross.
5. Wrap into a book with page 1 in front. Follow pages 1 to 8.



Pictures are simplified and not to scale. Look closely, wonder, and pause whenever you like.

The hammer has fallen back, but the string is not still. It quivers: back and forth, back and forth. That quick motion is vibration. "The tap is over," Little says, "but the moving isn't it" How can such a thin string sound so big?

With the pedal up, Little releases a middle key. A felt damper hushes its strings. Little sketches key, hammer, strings, wood. "The note wasn't hiding. We set things moving!" Big leaves room for one more why.

Little follows the same string to a wooden bridge. The bridge carries vibration into a broad soundboard. That wood vibrates too, moving more air. The air carries sound to listening ears. "The wood joins in!" Little says. A thin string has a roomy helper.

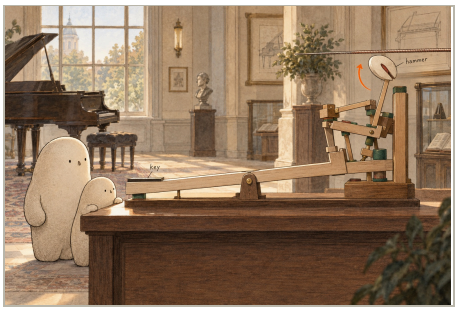
Two strings stretch across another model. One is longer. If everything else stays the same, the longer string vibrates more slowly and sounds lower. The shorter one sounds higher. "Different voices!" says Little. How high or low a note sounds is its pitch.

Give the note a path.

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Choose one picture from Little's search. Tell what moves and what happens next. Draw or write a clue that helped you understand the note.



A hidden tap



The wood joins in



Two lengths to notice

Draw one detail from your scene. Add an arrow or a label to show a movement or a difference.

key · hammer · string · vibration · bridge · soundboard · pitch

When a key moves, … What does the action help the hammer do?

The soundboard helps because … How does the broad wood help move air?

If only the string's length changes, … Which length could make the lower sound?

Read it together and point to the detail your words explain. If you are writing, check the first capital and the ending mark.